

Unit 15, Lesson 3: Probability



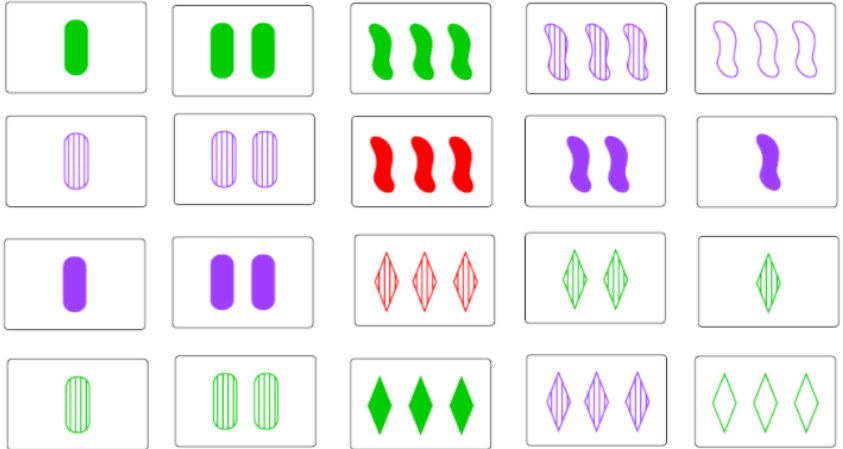
MA.7.DP.2.2 Given the probability of a chance event, interpret the likelihood of it occurring. Compare the probabilities of chance events.



- ☐ I can express the probability of a chance event using a fraction, percentage, or decimal.
- ☐ I can compare probabilities of chance events.



1. Some of the cards in the card game SET are shown.



- What do you notice?
- What do you wonder?



15.3.2 Exploration: Representing Probabilities

When performing simple experiments, all of the outcomes have the same likelihood of happening. For example, when flipping a fair coin, the sample space includes two possible outcomes, {heads, tails}. Each outcome has an equal chance of happening. Therefore, the chance of landing on heads is $\frac{1}{2}$ because this event represents 1 outcome out of 2 possible outcomes. Represented as a decimal or a percentage, the chance of landing on heads is 0.5 or 50%, respectively.

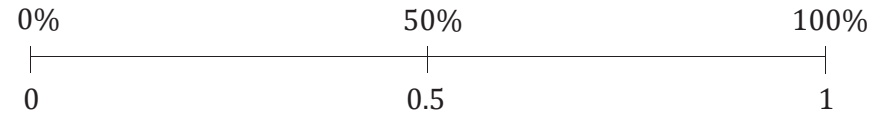
Work with your partner to complete the following.

1. From the percentages listed below, determine the percent chance of each event occurring. Each value is used once.

0% 16. $\bar{6}$ % 33. $\bar{3}$ % 50% 100%

Event	Percent Chance
a. When given three doors with only one containing a prize behind it, what is the chance of opening the prize-winning door?	
b. When reaching into a dark closet with five pairs of shoes, what is the chance of pulling out a left shoe?	
c. When prompted to randomly select a crayon from a group of red, blue, and yellow crayons, what is the chance you choose a purple crayon?	
d. When rolling a fair six-sided die, what is the chance of landing on 1?	
e. When flipping a coin, what is the chance of landing on either heads or tails?	

2. Plot and label a point to represent the percent chance of each event happening from the previous question. Use the letters labeling each event.



The **probability** of an event is a measure of the likelihood the event will occur. Probabilities are expressed using numbers from 0 to 1 as a **decimal, percentage, or fraction**.

3. Consider the event from question 1 that is least likely to occur.
 - a. Which event is the least likely to happen?
 - b. Describe, using sentences, the likelihood of this event happening.
 - c. Discuss with your partner whether or not it is possible to have a probability less than this.

4. Consider the event from question 1 that is most likely to occur.
- Which event is the most likely to occur?
 - Describe, in sentences, the likelihood of this event happening.
 - Discuss with your partner whether or not it is possible to have a probability greater than this.
5. Which event has the same likelihood of happening or not happening?



15.3.3 Bring It Together: Representing Probabilities

1. Complete the statements.

The probability of an event can be any number between

0 and 1. A probability of 0 means it is

- ☐ certain
- ☐ impossible

for the event to occur. A probability of 1 means it is

- ☐ certain
- ☐ impossible

the event will occur. The closer a

probability is to 1, the

- ☐ more
- ☐ less

likely it is to occur,

while the closer it is to 0, the

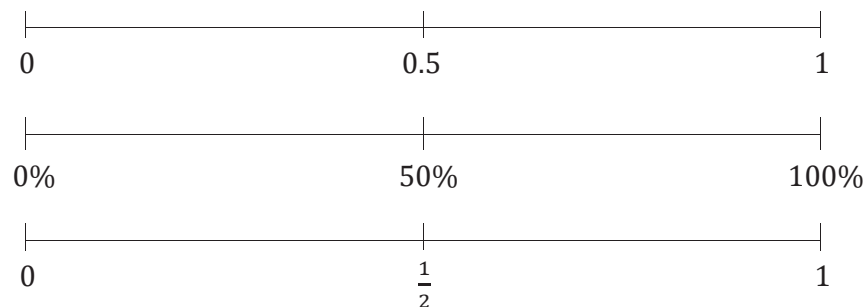
- ☐ more
- ☐ less

likely it is to

occur. A probability of 0.5 means the event is

- ☐ likely.
- ☐ unlikely.
- ☐ neither likely nor unlikely.

Probabilities can be represented by decimals, percentages, or fractions with equivalent values between 0 and 1.





15.3.4 Guided Instruction: Expressing Probabilities of Events



The **probability** of an event occurring, notated as $P(\text{event})$, is determined by dividing the number of outcomes in the event by the possible outcomes in the sample space.

$$P(\text{event}) = \frac{\text{number of outcomes in the event}}{\text{total possible outcomes}}$$

Harper is playing a game where she draws a lettered tile out of a bag without looking.

The notation $P(A)$ in this context means “the probability of pulling out a tile with the letter A on it.”



1. Find the probability of each event, represented as a fraction.

- a. $P(A)$
- b. $P(B)$
- c. $P(M)$
- d. $P(\text{vowel})$
- e. $P(\text{letter})$

2. Consider the probability of pulling a tile with a consonant on it.

- a. Complete the table to find the fraction, decimal, and percentage equivalents of the probability.

Event	Fraction	Decimal	Percent
$P(\text{consonant})$			

The probability of this event is a repeating decimal. When this occurs, the decimal or percentage representation is typically rounded.

- b. What is $P(\text{consonant})$ represented as a decimal rounded to the nearest hundredth?
- c. What is $P(\text{consonant})$ represented as a percentage rounded to the nearest tenth?



15.3.5 Try It: Expressing Probabilities of Events in Different Forms

1. The probability of rolling a 6 on a standard six-sided die is 16.67%, rounded to the nearest hundredth. Maisie says that this probability is greater than an event whose probability is 0.6 because $16.67 > 0.6$.

Explain whether you agree or disagree with Maisie's conclusion. Support your argument by showing your work.

2. The probabilities of five events are given.

A	B	C	D	E
40%	0.25	$\frac{5}{6}$	42.9%	$\frac{1}{5}$

- a. Order the values of the probabilities from least to greatest.

- b. Match the probabilities with the events described in the table.

Event	Probability
 Landing on green on the spinner shown	
Pulling a blue marble from a bag containing equal numbers of red, orange, yellow, green, and blue marbles	
Landing on a factor of 10 when rolling a fair ten-sided die	
Picking a card with a vowel on it from a set of cards with each letter in the word FLORIDA on it	
Landing on a number less than 6 on a standard six-sided die	

- c. Explain which event is the least likely to occur based on the probabilities.



15.3.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current level of understanding is related to each target.

1. I can express the probability of a chance event using a fraction, percentage, or decimal.

I need help. I'm getting there, but I understand this
I can't get started. I need more practice. and I feel confident.



2. I can compare probabilities of chance events.

I need help. I'm getting there, but I understand this
I can't get started. I need more practice. and I feel confident.



Unit 15, Lesson 4: Comparing Probabilities



Benchmark

MA.7.DP.2.2 Given the probability of a chance event, interpret the likelihood of it occurring. Compare the probabilities of chance events.



Learning Targets

- ☐ I can interpret the likelihood of a chance event occurring using probability.
- ☐ I can compare probabilities of chance events.